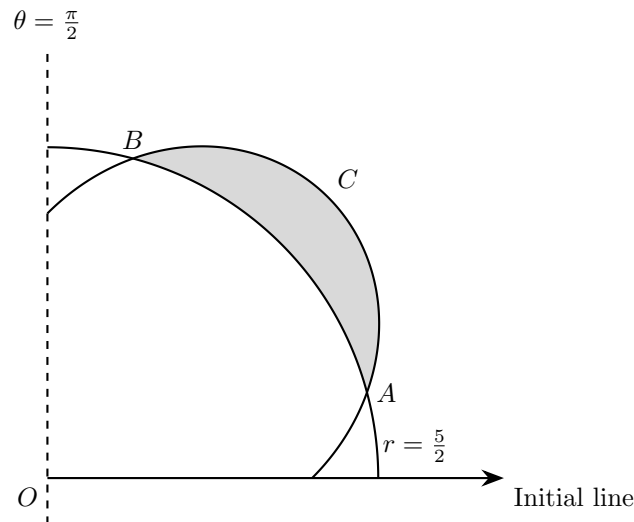


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1. The diagram shows part of the curve C with polar equation

$$r = 2 + \sin 2\theta, \quad 0 \leq \theta \leq \frac{\pi}{2}$$

and the circle

$$r = \frac{5}{2}$$

The curve and the circle intersect at the points A and B , as shown.

Find, correct to three significant figures, the area of the shaded region enclosed by C and the circle.

[9]

Solution

For the intersection points of the curve and the circle, set the two values of r equal:

$$2 + \sin 2\theta = \frac{5}{2}$$

So

$$\sin 2\theta = \frac{1}{2}$$

Since $0 \leq \theta \leq \frac{\pi}{2}$, we have $0 \leq 2\theta \leq \pi$, so the solutions are

$$2\theta = \frac{\pi}{6}, \quad \frac{5\pi}{6}$$

Hence

$$\theta = \frac{\pi}{12}, \quad \frac{5\pi}{12}$$

So the points of intersection are at $\theta = \frac{\pi}{12}$ and $\theta = \frac{5\pi}{12}$.

For $\frac{\pi}{12} \leq \theta \leq \frac{5\pi}{12}$, we have $\sin 2\theta \geq \frac{1}{2}$, so

$$2 + \sin 2\theta \geq \frac{5}{2}$$

Therefore the curve C lies outside the circle on this interval, and the required area is

$$A = \frac{1}{2} \int_{\pi/12}^{5\pi/12} \left[(2 + \sin 2\theta)^2 - \left(\frac{5}{2} \right)^2 \right] d\theta$$

Now expand the integrand:

$$\begin{aligned}
 A &= \frac{1}{2} \int_{\pi/12}^{5\pi/12} \left(4 + 4 \sin 2\theta + \sin^2 2\theta - \frac{25}{4} \right) d\theta \\
 &= \frac{1}{2} \int_{\pi/12}^{5\pi/12} \left(-\frac{9}{4} + 4 \sin 2\theta + \sin^2 2\theta \right) d\theta
 \end{aligned}$$

Use the identity

$$\sin^2 2\theta = \frac{1 - \cos 4\theta}{2}$$

Then

$$\begin{aligned}
 A &= \frac{1}{2} \int_{\pi/12}^{5\pi/12} \left(-\frac{9}{4} + 4 \sin 2\theta + \frac{1 - \cos 4\theta}{2} \right) d\theta \\
 &= \frac{1}{2} \int_{\pi/12}^{5\pi/12} \left(-\frac{7}{4} + 4 \sin 2\theta - \frac{1}{2} \cos 4\theta \right) d\theta
 \end{aligned}$$

Integrating,

$$A = \left[-\frac{7\theta}{8} - \cos 2\theta - \frac{1}{16} \sin 4\theta \right]_{\pi/12}^{5\pi/12}$$

Now substitute the limits:

$$A = \left(-\frac{35\pi}{96} - \cos \frac{5\pi}{6} - \frac{1}{16} \sin \frac{5\pi}{3} \right) - \left(-\frac{7\pi}{96} - \cos \frac{\pi}{6} - \frac{1}{16} \sin \frac{\pi}{3} \right)$$

Using

$$\cos \frac{5\pi}{6} = -\frac{\sqrt{3}}{2}, \quad \sin \frac{5\pi}{3} = -\frac{\sqrt{3}}{2}, \quad \cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}, \quad \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

gives

$$\begin{aligned}
 A &= \left(-\frac{35\pi}{96} + \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{32} \right) - \left(-\frac{7\pi}{96} - \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{32} \right) \\
 &= -\frac{28\pi}{96} + \frac{34\sqrt{3}}{32} \\
 &= -\frac{7\pi}{24} + \frac{17\sqrt{3}}{16}
 \end{aligned}$$

So the exact area is

$$A = \frac{17\sqrt{3}}{16} - \frac{7\pi}{24}$$

and numerically

$$A \approx 0.924$$

Therefore, the area of the shaded region is 0.924 square units, correct to three significant figures.

2. (a) A curve has polar equation $r = a \sin^2 3\theta$, where $a > 0$ and $0 \leq \theta < 2\pi$.

(i) Sketch the curve. [2]

(ii) Find, in exact form, the total area of the regions enclosed by the curve. [7]

Solution

(a) (i) Since

$$r = a \sin^2(3\theta)$$

we have $r \geq 0$ for all θ .

Also,

$$\sin^2(3(\theta + \frac{\pi}{3})) = \sin^2(3\theta + \pi) = \sin^2(3\theta)$$

so the curve has period $\frac{\pi}{3}$. Hence the part from $0 \leq \theta < \frac{\pi}{3}$ repeats 6 times as θ goes from 0 to 2π .

The curve passes through the pole when

$$\sin^2(3\theta) = 0 \Rightarrow 3\theta = n\pi \Rightarrow \theta = \frac{n\pi}{3}$$

for $n = 0, 1, 2, 3, 4, 5$.

The maximum value of r is a , and this occurs when

$$\sin^2(3\theta) = 1 \Rightarrow 3\theta = \frac{\pi}{2} + n\pi$$

so

$$\theta = \frac{\pi}{6} + \frac{n\pi}{3}$$

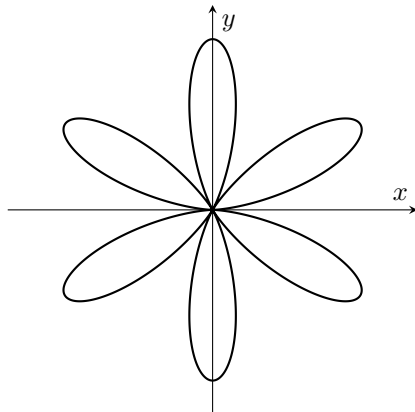
for $n = 0, 1, 2, 3, 4, 5$.

So the curve is a 6-petalled rose, with petals centred on

$$\theta = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$$

and each petal has maximum radius a .

A sketch is shown below.



Hence the sketch is a 6-petalled rose curve.

(ii) The polar area formula is

$$A = \frac{1}{2} \int r^2 d\theta$$

One petal is traced out as θ goes from 0 to $\frac{\pi}{3}$, since $r = 0$ at both endpoints and $r > 0$ between them.

So the area of one petal is

$$A_1 = \frac{1}{2} \int_0^{\pi/3} a^2 \sin^4(3\theta) \, d\theta$$

Use

$$\sin^4 x = \left(\frac{1 - \cos 2x}{2} \right)^2 = \frac{1 - 2 \cos 2x + \cos^2 2x}{4} = \frac{1 - 2 \cos 2x + \frac{1 + \cos 4x}{2}}{4} = \frac{3 - 4 \cos 2x + \cos 4x}{8}$$

Therefore

$$\sin^4(3\theta) = \frac{3 - 4 \cos 6\theta + \cos 12\theta}{8}$$

Substitute into the integral:

$$\begin{aligned} A_1 &= \frac{a^2}{2} \int_0^{\pi/3} \frac{3 - 4 \cos 6\theta + \cos 12\theta}{8} \, d\theta \\ &= \frac{a^2}{16} \int_0^{\pi/3} (3 - 4 \cos 6\theta + \cos 12\theta) \, d\theta \\ &= \frac{a^2}{16} \left[3\theta - \frac{4}{6} \sin 6\theta + \frac{1}{12} \sin 12\theta \right]_0^{\pi/3} \end{aligned}$$

Now evaluate:

$$\sin \left(6 \cdot \frac{\pi}{3} \right) = \sin 2\pi = 0, \quad \sin \left(12 \cdot \frac{\pi}{3} \right) = \sin 4\pi = 0$$

Hence

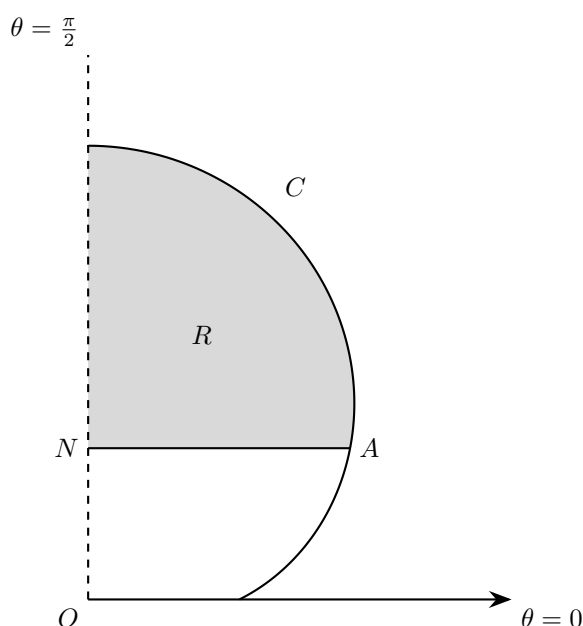
$$\begin{aligned} A_1 &= \frac{a^2}{16} \left[3 \cdot \frac{\pi}{3} - 0 + 0 \right] \\ &= \frac{\pi a^2}{16} \end{aligned}$$

There are 6 identical petals, so the total area is

$$\begin{aligned} A_{\text{total}} &= 6 \cdot \frac{\pi a^2}{16} \\ &= \frac{3\pi a^2}{8} \end{aligned}$$

Therefore the total area enclosed by the curve is

$$\frac{3\pi a^2}{8}$$



3. The curve C shown in the diagram above has polar equation

$$r = 1 + 2 \sin \theta, \quad 0 \leq \theta \leq \frac{\pi}{2}$$

The point A on C has y -coordinate 1.

The point N lies on the line $\theta = \frac{\pi}{2}$ and AN is perpendicular to the line $\theta = \frac{\pi}{2}$.

The finite region R , shown shaded in the diagram above, is bounded by the curve C , the line $\theta = \frac{\pi}{2}$ and the line AN .

Find the exact area of the shaded region R .

[9]

Solution

For the curve

$$r = 1 + 2 \sin \theta, \quad 0 \leq \theta \leq \frac{\pi}{2}$$

the point A has y -coordinate 1.

In polar coordinates,

$$y = r \sin \theta$$

so at A ,

$$(1 + 2 \sin \theta) \sin \theta = 1$$

Hence

$$\begin{aligned} 2 \sin^2 \theta + \sin \theta - 1 &= 0 \\ (2 \sin \theta - 1)(\sin \theta + 1) &= 0 \end{aligned}$$

Since $0 \leq \theta \leq \frac{\pi}{2}$, we take

$$\sin \theta = \frac{1}{2}$$

so

$$\theta_A = \frac{\pi}{6}$$

Then

$$r_A = 1 + 2 \sin \frac{\pi}{6} = 1 + 2 \left(\frac{1}{2} \right) = 2$$

Therefore

$$A = \left(2, \frac{\pi}{6}\right)$$

and in Cartesian coordinates

$$x = 2 \cos \frac{\pi}{6} = \sqrt{3}, \quad y = 2 \sin \frac{\pi}{6} = 1$$

so

$$A = (\sqrt{3}, 1)$$

The line $\theta = \frac{\pi}{2}$ is the y -axis, so the line AN , perpendicular to it, is horizontal. Since it passes through A , its equation is

$$y = 1$$

Now consider the shaded region R . For angles

$$\frac{\pi}{6} \leq \theta \leq \frac{\pi}{2}$$

the outer boundary is the curve

$$r = 1 + 2 \sin \theta$$

and the inner boundary is the line $y = 1$.

Using $y = r \sin \theta$, the line $y = 1$ becomes

$$r \sin \theta = 1$$

so

$$r = \frac{1}{\sin \theta} = \operatorname{cosec} \theta$$

Using the polar area formula,

$$\text{Area}(R) = \frac{1}{2} \int_{\pi/6}^{\pi/2} \left[(1 + 2 \sin \theta)^2 - \operatorname{cosec}^2 \theta \right] d\theta$$

Expand the bracket:

$$\text{Area}(R) = \frac{1}{2} \int_{\pi/6}^{\pi/2} (1 + 4 \sin \theta + 4 \sin^2 \theta - \operatorname{cosec}^2 \theta) d\theta$$

Now integrate term by term. Using

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

we get

$$\begin{aligned} \int (1 + 4 \sin \theta + 4 \sin^2 \theta - \operatorname{cosec}^2 \theta) d\theta &= \int (1 + 4 \sin \theta + 2 - 2 \cos 2\theta - \operatorname{cosec}^2 \theta) d\theta \\ &= 3\theta - 4 \cos \theta - \sin 2\theta + \cot \theta \end{aligned}$$

So

$$\text{Area}(R) = \frac{1}{2} [3\theta - 4 \cos \theta - \sin 2\theta + \cot \theta]_{\pi/6}^{\pi/2}$$

Now substitute the limits.

At $\theta = \frac{\pi}{2}$:

$$3 \left(\frac{\pi}{2} \right) - 4 \cos \frac{\pi}{2} - \sin \pi + \cot \frac{\pi}{2} = \frac{3\pi}{2}$$

At $\theta = \frac{\pi}{6}$:

$$\begin{aligned} 3 \left(\frac{\pi}{6} \right) - 4 \cos \frac{\pi}{6} - \sin \frac{\pi}{3} + \cot \frac{\pi}{6} &= \frac{\pi}{2} - 4 \left(\frac{\sqrt{3}}{2} \right) - \frac{\sqrt{3}}{2} + \sqrt{3} \\ &= \frac{\pi}{2} - 2\sqrt{3} - \frac{\sqrt{3}}{2} + \sqrt{3} \\ &= \frac{\pi}{2} - \frac{3\sqrt{3}}{2} \end{aligned}$$

Therefore

$$\begin{aligned}\text{Area}(R) &= \frac{1}{2} \left(\frac{3\pi}{2} - \left(\frac{\pi}{2} - \frac{3\sqrt{3}}{2} \right) \right) \\ &= \frac{1}{2} \left(\pi + \frac{3\sqrt{3}}{2} \right) \\ &= \frac{\pi}{2} + \frac{3\sqrt{3}}{4}\end{aligned}$$

Hence the exact area of the shaded region is

$$\frac{\pi}{2} + \frac{3\sqrt{3}}{4}$$

4. The curve C has polar equation

$$r^2 = \frac{\pi^2 - (\theta - \pi)^2}{1 + (\theta - \pi)^2} \quad 0 \leq \theta \leq 2\pi$$

(a) Sketch C and state the polar coordinates of the point on C furthest from the pole. [4]

(b) Using the substitution $u = \theta - \pi$, or otherwise, find the exact area enclosed by C . [6]

Solution

(a) Let

$$u = (\theta - \pi)^2$$

Then

$$r^2 = \frac{\pi^2 - u}{1 + u}, \quad 0 \leq u \leq \pi^2$$

Differentiate with respect to u :

$$\frac{d(r^2)}{du} = \frac{-(1+u) - (\pi^2 - u)}{(1+u)^2} = -\frac{\pi^2 + 1}{(1+u)^2} < 0$$

So r^2 decreases as u increases, so its maximum occurs when $u = 0$.

Hence

$$(\theta - \pi)^2 = 0 \implies \theta = \pi$$

and then

$$r^2 = \frac{\pi^2}{1} = \pi^2$$

Since we take $r > 0$, this gives

$$r = \pi$$

So the point on C furthest from the pole is

$$(\pi, \pi)$$

For the sketch, note that when $\theta = 0$ or $\theta = 2\pi$,

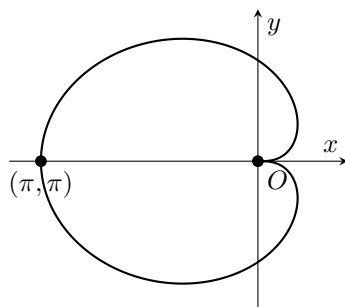
$$r^2 = \frac{\pi^2 - \pi^2}{1 + \pi^2} = 0$$

so the curve passes through the pole.

Also r depends only on $(\theta - \pi)^2$, so

$$r(2\pi - \theta) = r(\theta)$$

Hence the curve is symmetric about the polar axis. Therefore C is a single closed loop through the pole, with greatest extent on the negative x -axis.



(b) The area enclosed by a polar curve is

$$A = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta$$

So here

$$A = \frac{1}{2} \int_0^{2\pi} \frac{\pi^2 - (\theta - \pi)^2}{1 + (\theta - \pi)^2} d\theta$$

Use the substitution

$$u = \theta - \pi$$

Then $du = d\theta$, and the limits change as follows:

$$\theta = 0 \implies u = -\pi, \quad \theta = 2\pi \implies u = \pi$$

Hence

$$A = \frac{1}{2} \int_{-\pi}^{\pi} \frac{\pi^2 - u^2}{1 + u^2} du$$

Now simplify the integrand:

$$\begin{aligned} \frac{\pi^2 - u^2}{1 + u^2} &= \frac{-(1 + u^2) + (\pi^2 + 1)}{1 + u^2} \\ &= -1 + \frac{\pi^2 + 1}{1 + u^2} \end{aligned}$$

Therefore

$$\begin{aligned} A &= \frac{1}{2} \int_{-\pi}^{\pi} \left(-1 + \frac{\pi^2 + 1}{1 + u^2} \right) du \\ &= \frac{1}{2} [-u + (\pi^2 + 1) \arctan u]_{-\pi}^{\pi} \end{aligned}$$

using

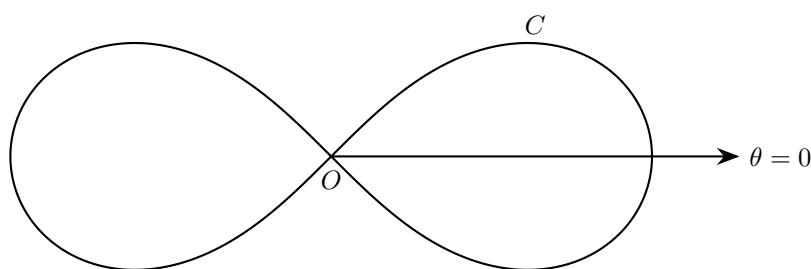
$$\int \frac{1}{1 + u^2} du = \arctan u + c$$

Now substitute the limits:

$$\begin{aligned} A &= \frac{1}{2} \left[(-\pi + (\pi^2 + 1) \arctan \pi) - (\pi - (\pi^2 + 1) \arctan \pi) \right] \\ &= \frac{1}{2} \left[-2\pi + 2(\pi^2 + 1) \arctan \pi \right] \\ &= (\pi^2 + 1) \arctan(\pi) - \pi \end{aligned}$$

So the exact area enclosed by C is

$$(\pi^2 + 1) \arctan(\pi) - \pi$$



5. The diagram below shows the curve C with polar equation $r^2 = 8 \cos 2\theta$.

(a) Show that a cartesian equation of C is

$$(x^2 + y^2)^2 = 8(x^2 - y^2) \quad [4]$$

(b) Show that the coordinate axes are lines of symmetry of C . [2]

(c) Find the exact area enclosed by each loop of C . [5]

Solution

(a) Using

$$x = r \cos \theta, \quad y = r \sin \theta$$

we have

$$r^2 = x^2 + y^2$$

Also,

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

and since

$$\cos \theta = \frac{x}{r}, \quad \sin \theta = \frac{y}{r}$$

it follows that

$$\begin{aligned} \cos 2\theta &= \left(\frac{x}{r}\right)^2 - \left(\frac{y}{r}\right)^2 \\ &= \frac{x^2 - y^2}{r^2} \end{aligned}$$

Now substitute into the polar equation $r^2 = 8 \cos 2\theta$:

$$r^2 = 8 \cdot \frac{x^2 - y^2}{r^2}$$

and since $r^2 = x^2 + y^2$,

$$x^2 + y^2 = \frac{8(x^2 - y^2)}{x^2 + y^2}$$

Multiplying both sides by $x^2 + y^2$ gives

$$(x^2 + y^2)^2 = 8(x^2 - y^2)$$

So a cartesian equation of C is

$$(x^2 + y^2)^2 = 8(x^2 - y^2)$$

(b) To test symmetry in the x -axis, replace y by $-y$:

$$\begin{aligned}(x^2 + (-y)^2)^2 &= 8(x^2 - (-y)^2) \\ (x^2 + y^2)^2 &= 8(x^2 - y^2)\end{aligned}$$

which is unchanged, so the curve is symmetric about the x -axis.

To test symmetry in the y -axis, replace x by $-x$:

$$\begin{aligned}((-x)^2 + y^2)^2 &= 8((-x)^2 - y^2) \\ (x^2 + y^2)^2 &= 8(x^2 - y^2)\end{aligned}$$

which is also unchanged, so the curve is symmetric about the y -axis.

Hence the coordinate axes are lines of symmetry of C .

(c) For a polar curve, the area enclosed is

$$A = \frac{1}{2} \int r^2 \, d\theta$$

Here

$$r^2 = 8 \cos 2\theta$$

One loop is traced when $r^2 \geq 0$ and θ runs from $-\frac{\pi}{4}$ to $\frac{\pi}{4}$, since at the endpoints

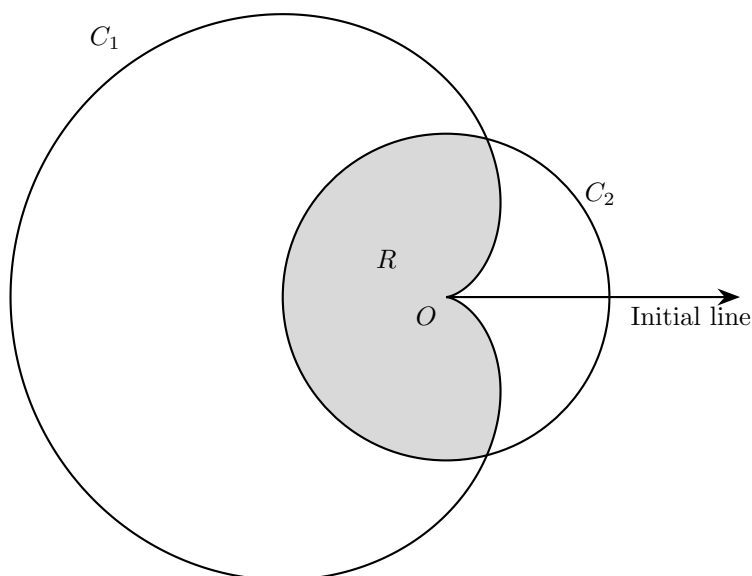
$$\cos 2\theta = 0$$

so the curve passes through the pole.

Therefore the area of one loop is

$$\begin{aligned}A &= \frac{1}{2} \int_{-\pi/4}^{\pi/4} 8 \cos 2\theta \, d\theta \\ &= 4 \int_{-\pi/4}^{\pi/4} \cos 2\theta \, d\theta \\ &= 4 \left[\frac{1}{2} \sin 2\theta \right]_{-\pi/4}^{\pi/4} \\ &= 2 [\sin 2\theta]_{-\pi/4}^{\pi/4} \\ &= 2 \left(\sin \frac{\pi}{2} - \sin \left(-\frac{\pi}{2} \right) \right) \\ &= 2(1 - (-1)) \\ &= 4\end{aligned}$$

So the exact area enclosed by each loop is



6. The diagram above shows the curve C_1 with polar equation $r = 2a(1 - \cos \theta)$, $0 \leq \theta \leq 2\pi$, and the circle C_2 with polar equation $r = a$, $0 \leq \theta \leq 2\pi$, where a is a positive constant.

(a) Find, in terms of a , the polar coordinates of the points where C_1 meets C_2 .

[3]

The common region enclosed by C_1 and C_2 is shaded in the diagram above.

(b) Find the area of the shaded region R , giving your answer in the form

$$\frac{1}{12}a^2 (p\pi + q\sqrt{3})$$

where p and q are integers to be found.

[7]

Solution

(a) At the points of intersection, the two values of r are equal, so

$$2a(1 - \cos \theta) = a$$

Since $a > 0$, divide through by a :

$$2(1 - \cos \theta) = 1$$

$$2 - 2\cos \theta = 1$$

$$\cos \theta = \frac{1}{2}$$

For $0 \leq \theta \leq 2\pi$, this gives

$$\theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

Substituting into either curve gives

$$r = a$$

So the points where C_1 meets C_2 are

$$(a, \frac{\pi}{3}) \quad \text{and} \quad (a, \frac{5\pi}{3})$$

(b) For the common region, the boundary is given by the smaller of the two radii.

Compare the curves:

$$2a(1 - \cos \theta) \leq a$$

$$2(1 - \cos \theta) \leq 1$$

$$\cos \theta \geq \frac{1}{2}$$

So:

$$0 \leq \theta \leq \frac{\pi}{3} \quad \text{and} \quad \frac{5\pi}{3} \leq \theta \leq 2\pi$$

give the cardioid $r = 2a(1 - \cos \theta)$ as the inner curve, and

$$\frac{\pi}{3} \leq \theta \leq \frac{5\pi}{3}$$

gives the circle $r = a$ as the inner curve.

Using the polar area formula

$$A = \frac{1}{2} \int r^2 d\theta$$

the shaded area is

$$A = \frac{1}{2} \left[\int_0^{\pi/3} (2a(1 - \cos \theta))^2 d\theta + \int_{\pi/3}^{5\pi/3} a^2 d\theta + \int_{5\pi/3}^{2\pi} (2a(1 - \cos \theta))^2 d\theta \right]$$

By symmetry, the first and third integrals are equal, so

$$\begin{aligned} A &= \int_0^{\pi/3} (2a(1 - \cos \theta))^2 d\theta + \frac{1}{2} \int_{\pi/3}^{5\pi/3} a^2 d\theta \\ &= 4a^2 \int_0^{\pi/3} (1 - 2\cos \theta + \cos^2 \theta) d\theta + \frac{1}{2} a^2 \left(\frac{5\pi}{3} - \frac{\pi}{3} \right) \end{aligned}$$

Now use

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

So

$$\begin{aligned} A &= 4a^2 \int_0^{\pi/3} \left(1 - 2\cos \theta + \frac{1 + \cos 2\theta}{2} \right) d\theta + \frac{1}{2} a^2 \cdot \frac{4\pi}{3} \\ &= 4a^2 \int_0^{\pi/3} \left(\frac{3}{2} - 2\cos \theta + \frac{1}{2} \cos 2\theta \right) d\theta + \frac{2\pi a^2}{3} \end{aligned}$$

Integrating,

$$\begin{aligned} A &= 4a^2 \left[\frac{3}{2} \theta - 2 \sin \theta + \frac{1}{4} \sin 2\theta \right]_0^{\pi/3} + \frac{2\pi a^2}{3} \\ &= 4a^2 \left(\frac{3}{2} \cdot \frac{\pi}{3} - 2 \sin \frac{\pi}{3} + \frac{1}{4} \sin \frac{2\pi}{3} \right) + \frac{2\pi a^2}{3} \\ &= 4a^2 \left(\frac{\pi}{2} - 2 \cdot \frac{\sqrt{3}}{2} + \frac{1}{4} \cdot \frac{\sqrt{3}}{2} \right) + \frac{2\pi a^2}{3} \\ &= 4a^2 \left(\frac{\pi}{2} - \sqrt{3} + \frac{\sqrt{3}}{8} \right) + \frac{2\pi a^2}{3} \\ &= 4a^2 \left(\frac{\pi}{2} - \frac{7\sqrt{3}}{8} \right) + \frac{2\pi a^2}{3} \\ &= 2\pi a^2 - \frac{7\sqrt{3}}{2} a^2 + \frac{2\pi a^2}{3} \\ &= \frac{8\pi a^2}{3} - \frac{7\sqrt{3} a^2}{2} \end{aligned}$$

Putting this over a denominator of 12,

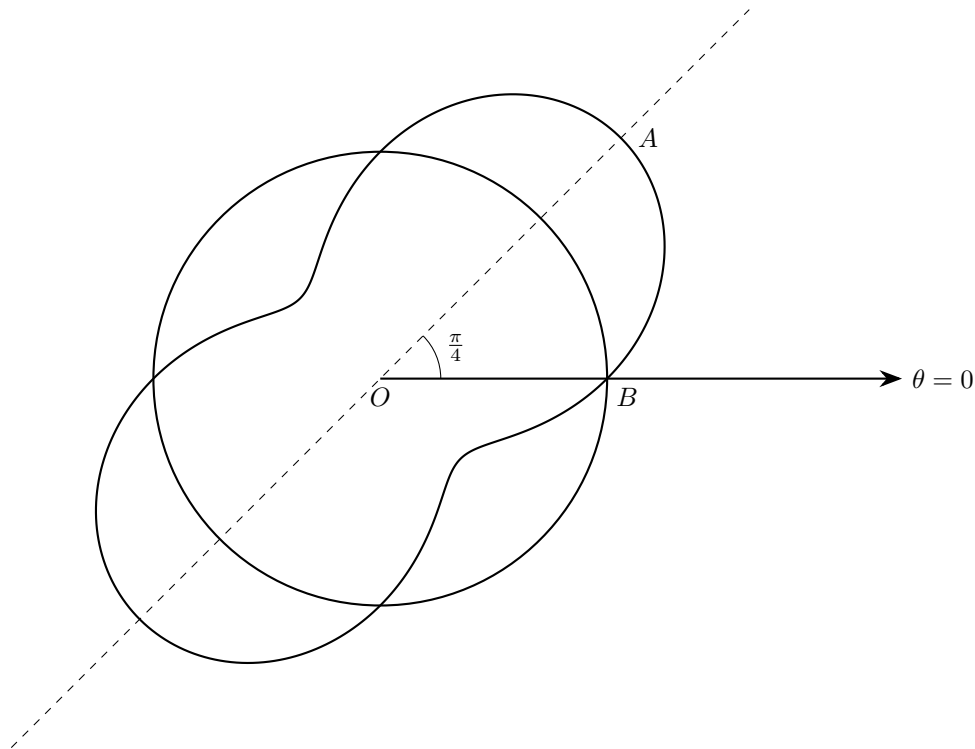
$$A = \frac{1}{12}a^2(32\pi - 42\sqrt{3})$$

Hence the shaded area is

$$\frac{1}{12}a^2(32\pi - 42\sqrt{3})$$

so

$$p = 32, \quad q = -42$$



7. The diagram above shows the curve C with polar equation $r = a(2 + \sin 2\theta)$ and the circle $r = 2a$ for $-\pi \leq \theta \leq \pi$, where a is a positive constant.

- (a) Write down the polar coordinates of the points A and B . [2]
- (b) Explain why C is symmetrical about the line $\theta = \frac{\pi}{4}$. [2]
- (c) Find, in terms of a , the exact total area of the regions that lie inside C and outside the circle. [5]

Solution

(a) For point A , the diagram shows $\theta = \frac{\pi}{4}$. Then

$$r = a \left(2 + \sin \frac{\pi}{2} \right) = a(2 + 1) = 3a$$

so

$$A = \left(3a, \frac{\pi}{4} \right)$$

For point B , $\theta = 0$, so

$$r = a(2 + \sin 0) = 2a$$

hence

$$B = (2a, 0)$$

(b) A curve is symmetrical about the line $\theta = \frac{\pi}{4}$ if replacing θ by

$$\frac{\pi}{2} - \theta$$

leaves the equation unchanged, since reflection in the line $\theta = \frac{\pi}{4}$ sends θ to $\frac{\pi}{2} - \theta$.

Here,

$$r = a \left(2 + \sin 2 \left(\frac{\pi}{2} - \theta \right) \right) = a(2 + \sin(\pi - 2\theta))$$

and since $\sin(\pi - 2\theta) = \sin 2\theta$,

$$r = a(2 + \sin 2\theta)$$

So the equation is unchanged, and therefore C is symmetrical about the line $\theta = \frac{\pi}{4}$.

(c) The region is inside C and outside the circle $r = 2a$, so we need

$$a(2 + \sin 2\theta) \geq 2a$$

Since $a > 0$, this gives

$$2 + \sin 2\theta \geq 2$$

$$\sin 2\theta \geq 0$$

On $-\pi \leq \theta \leq \pi$, this happens on

$$-\pi \leq \theta \leq -\frac{\pi}{2} \quad \text{and} \quad 0 \leq \theta \leq \frac{\pi}{2}$$

so there are two equal regions.

Find the area of one region, for $0 \leq \theta \leq \frac{\pi}{2}$.

Using the polar area formula,

$$\text{Area} = \frac{1}{2} \int r^2 d\theta$$

the required area for one region is

$$A_1 = \frac{1}{2} \int_0^{\pi/2} (a^2(2 + \sin 2\theta)^2 - (2a)^2) d\theta$$

$$A_1 = \frac{a^2}{2} \int_0^{\pi/2} ((2 + \sin 2\theta)^2 - 4) d\theta$$

Expand the bracket:

$$(2 + \sin 2\theta)^2 - 4 = 4 + 4 \sin 2\theta + \sin^2 2\theta - 4 = 4 \sin 2\theta + \sin^2 2\theta$$

So

$$A_1 = \frac{a^2}{2} \int_0^{\pi/2} (4 \sin 2\theta + \sin^2 2\theta) d\theta$$

Now evaluate each part:

$$\int_0^{\pi/2} 4 \sin 2\theta d\theta = [-2 \cos 2\theta]_0^{\pi/2} = 2 - (-2) = 4$$

Also,

$$\sin^2 2\theta = \frac{1 - \cos 4\theta}{2}$$

so

$$\int_0^{\pi/2} \sin^2 2\theta d\theta = \int_0^{\pi/2} \frac{1 - \cos 4\theta}{2} d\theta = \left[\frac{\theta}{2} - \frac{\sin 4\theta}{8} \right]_0^{\pi/2} = \frac{\pi}{4}$$

Therefore

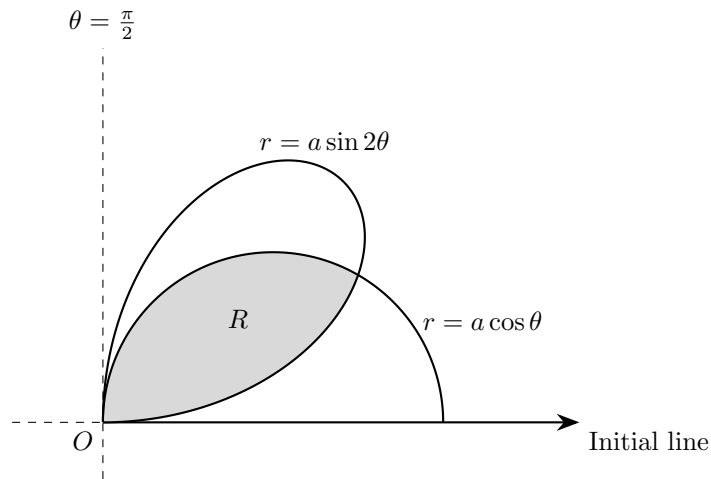
$$A_1 = \frac{a^2}{2} \left(4 + \frac{\pi}{4} \right) = a^2 \left(2 + \frac{\pi}{8} \right)$$

There are two such regions, so the total area is

$$2A_1 = 2a^2 \left(2 + \frac{\pi}{8} \right) = a^2 \left(4 + \frac{\pi}{4} \right) = \frac{a^2(16 + \pi)}{4}$$

Hence the exact total area is

$$\frac{a^2(16 + \pi)}{4}$$



8. In the first quadrant, the curve $r = a \sin 2\theta$ intersects the circle $r = a \cos \theta$ at the pole and at one other point.

The region R shown consists of the points that lie inside both curves.

Find the exact area of R in terms of a .

[8]

Solution

For the points of intersection, set the two expressions for r equal:

$$a \sin 2\theta = a \cos \theta$$

Since $a \neq 0$,

$$\sin 2\theta = \cos \theta$$

Using $\sin 2\theta = 2 \sin \theta \cos \theta$,

$$2 \sin \theta \cos \theta = \cos \theta$$

$$\cos \theta (2 \sin \theta - 1) = 0$$

In the first quadrant, $0 \leq \theta \leq \frac{\pi}{2}$, so

$$\cos \theta = 0 \Rightarrow \theta = \frac{\pi}{2}$$

or

$$2 \sin \theta - 1 = 0 \Rightarrow \sin \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}$$

At $\theta = \frac{\pi}{2}$,

$$r = a \cos \frac{\pi}{2} = 0$$

so this is the pole.

At the other intersection,

$$r = a \cos \frac{\pi}{6} = \frac{a\sqrt{3}}{2}$$

So the non-pole intersection is at

$$(r, \theta) = \left(\frac{a\sqrt{3}}{2}, \frac{\pi}{6} \right)$$

Now find which curve forms the boundary of the region inside both curves.

For $0 \leq \theta \leq \frac{\pi}{6}$,

$$a \sin 2\theta \leq a \cos \theta$$

so the smaller radius is $r = a \sin 2\theta$.

For $\frac{\pi}{6} \leq \theta \leq \frac{\pi}{2}$,

$$a \cos \theta \leq a \sin 2\theta$$

so the smaller radius is $r = a \cos \theta$.
Hence, using the polar area formula

$$\text{Area} = \frac{1}{2} \int r^2 d\theta$$

the required area is

$$\text{Area}(R) = \frac{1}{2} a^2 \left(\int_0^{\pi/6} \sin^2 2\theta d\theta + \int_{\pi/6}^{\pi/2} \cos^2 \theta d\theta \right)$$

Now evaluate the integrals.

First, use

$$\sin^2 2\theta = \frac{1 - \cos 4\theta}{2}$$

Then

$$\begin{aligned} \int_0^{\pi/6} \sin^2 2\theta d\theta &= \int_0^{\pi/6} \frac{1 - \cos 4\theta}{2} d\theta \\ &= \left[\frac{\theta}{2} - \frac{\sin 4\theta}{8} \right]_0^{\pi/6} \\ &= \frac{\pi}{12} - \frac{\sin(2\pi/3)}{8} \\ &= \frac{\pi}{12} - \frac{\sqrt{3}}{16} \end{aligned}$$

Next, use

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

Then

$$\begin{aligned} \int_{\pi/6}^{\pi/2} \cos^2 \theta d\theta &= \int_{\pi/6}^{\pi/2} \frac{1 + \cos 2\theta}{2} d\theta \\ &= \left[\frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_{\pi/6}^{\pi/2} \\ &= \left(\frac{\pi}{4} + 0 \right) - \left(\frac{\pi}{12} + \frac{\sin(\pi/3)}{4} \right) \\ &= \frac{\pi}{6} - \frac{\sqrt{3}}{8} \end{aligned}$$

Substitute back:

$$\begin{aligned} \text{Area}(R) &= \frac{1}{2} a^2 \left(\frac{\pi}{12} - \frac{\sqrt{3}}{16} + \frac{\pi}{6} - \frac{\sqrt{3}}{8} \right) \\ &= \frac{1}{2} a^2 \left(\frac{\pi}{4} - \frac{3\sqrt{3}}{16} \right) \\ &= \frac{a^2(4\pi - 3\sqrt{3})}{32} \end{aligned}$$

Hence the exact area of R is

$$\frac{a^2(4\pi - 3\sqrt{3})}{32}$$

9. (a) On the same diagram, sketch the curve C_1 with polar equation

$$r = 2(1 + \cos \theta), \quad 0 \leq \theta \leq 2\pi$$

and the curve C_2 with polar equation $\theta = \frac{2\pi}{3}$. [3]

- (b) Find the area of the smaller region bounded by C_1 and C_2 . [6]

Solution

- (a) For C_1 ,

$$r = 2(1 + \cos \theta)$$

Since this depends on $\cos \theta$, the curve is symmetric about the initial line.

Key values are

$$\theta = 0 \implies r = 4, \quad \theta = \pi \implies r = 0, \quad \theta = \frac{\pi}{2}, \frac{3\pi}{2} \implies r = 2$$

So C_1 is a cardioid with maximum $r = 4$ on the initial line and a cusp at the pole.

For C_2 ,

$$\theta = \frac{2\pi}{3}$$

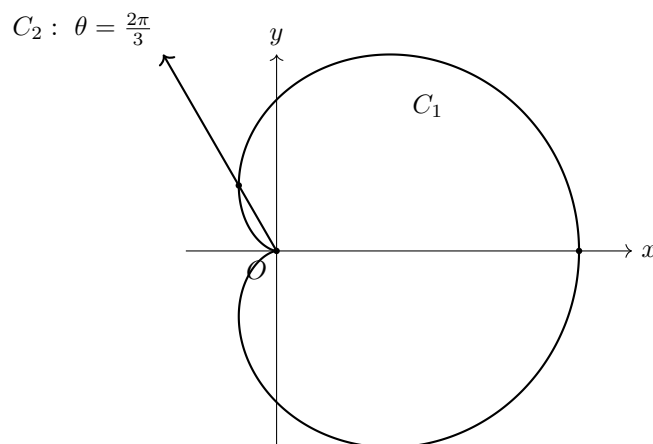
is the ray through the pole making an angle 120° with the initial line.

The other intersection of the two curves is found by substituting $\theta = \frac{2\pi}{3}$ into C_1 :

$$r = 2 \left(1 + \cos \frac{2\pi}{3} \right) = 2 \left(1 - \frac{1}{2} \right) = 1$$

So the curves meet at the pole and at $r = 1$, $\theta = \frac{2\pi}{3}$.

A suitable sketch is:



- (b) The smaller region is the part inside the cardioid between

$$\theta = \frac{2\pi}{3} \quad \text{and} \quad \theta = \pi$$

Using the polar area formula

$$A = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta$$

we get

$$A = \frac{1}{2} \int_{2\pi/3}^{\pi} [2(1 + \cos \theta)]^2 d\theta$$

Now simplify the integrand:

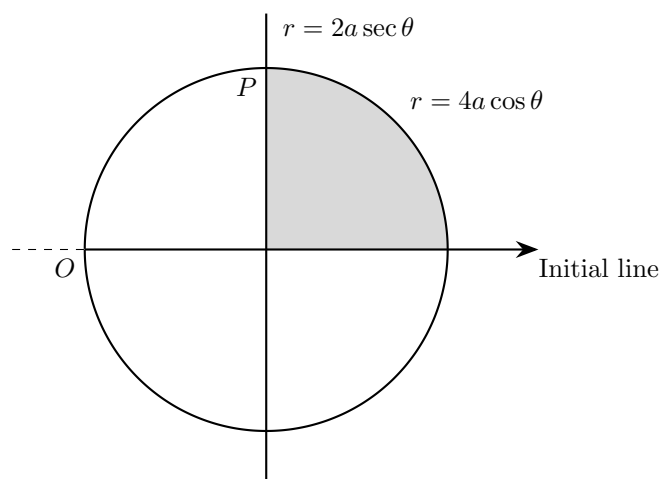
$$\begin{aligned} A &= 2 \int_{2\pi/3}^{\pi} (1 + 2 \cos \theta + \cos^2 \theta) \, d\theta \\ &= 2 \int_{2\pi/3}^{\pi} \left(1 + 2 \cos \theta + \frac{1 + \cos 2\theta}{2} \right) \, d\theta \\ &= \int_{2\pi/3}^{\pi} (3 + 4 \cos \theta + \cos 2\theta) \, d\theta \end{aligned}$$

Integrating,

$$\begin{aligned} A &= \left[3\theta + 4 \sin \theta + \frac{1}{2} \sin 2\theta \right]_{2\pi/3}^{\pi} \\ &= \left(3\pi + 4 \sin \pi + \frac{1}{2} \sin 2\pi \right) - \left(3 \cdot \frac{2\pi}{3} + 4 \sin \frac{2\pi}{3} + \frac{1}{2} \sin \frac{4\pi}{3} \right) \\ &= 3\pi - \left(2\pi + 4 \cdot \frac{\sqrt{3}}{2} + \frac{1}{2} \cdot \left(-\frac{\sqrt{3}}{2} \right) \right) \\ &= \pi - 2\sqrt{3} + \frac{\sqrt{3}}{4} \\ &= \pi - \frac{7\sqrt{3}}{4} \end{aligned}$$

So the area of the smaller region is

$$\pi - \frac{7\sqrt{3}}{4} \approx 0.111$$



10. The diagram shows a sketch of the curves with polar equations

$$r = 4a \cos \theta \text{ and } r = 2a \sec \theta, \quad a > 0$$

- (a) Find the polar coordinates of the point of intersection P of the two curves in the first quadrant. [4]
- (b) Find the area, shaded in the figure, bounded by the two curves and by the initial line $\theta = 0$, giving your answer in terms of a and π . [6]

Solution

(a) At a point of intersection, the two expressions for r are equal:

$$4a \cos \theta = 2a \sec \theta$$

Since $a > 0$, divide by $2a$:

$$2 \cos \theta = \sec \theta = \frac{1}{\cos \theta}$$

So

$$\begin{aligned} 2 \cos^2 \theta &= 1 \\ \cos^2 \theta &= \frac{1}{2} \end{aligned}$$

In the first quadrant, $\cos \theta > 0$, so

$$\cos \theta = \frac{1}{\sqrt{2}}$$

hence

$$\theta = \frac{\pi}{4}$$

Now find r :

$$r = 4a \cos \frac{\pi}{4} = 4a \cdot \frac{1}{\sqrt{2}} = 2\sqrt{2}a$$

So the polar coordinates of P are

$$\left(2\sqrt{2}a, \frac{\pi}{4} \right)$$

(b) The required shaded area is between the curves from $\theta = 0$ to the intersection at $\theta = \frac{\pi}{4}$.

For $0 \leq \theta \leq \frac{\pi}{4}$, the outer curve is

$$r = 4a \cos \theta$$

and the inner curve is

$$r = 2a \sec \theta$$

So, using

$$\text{Area} = \frac{1}{2} \int_{\alpha}^{\beta} (r_{\text{outer}}^2 - r_{\text{inner}}^2) \, d\theta$$

we get

$$A = \frac{1}{2} \int_0^{\pi/4} ((4a \cos \theta)^2 - (2a \sec \theta)^2) \, d\theta$$

Simplify:

$$A = \frac{1}{2} \int_0^{\pi/4} (16a^2 \cos^2 \theta - 4a^2 \sec^2 \theta) \, d\theta$$

$$A = 8a^2 \int_0^{\pi/4} \cos^2 \theta \, d\theta - 2a^2 \int_0^{\pi/4} \sec^2 \theta \, d\theta$$

Use

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

so

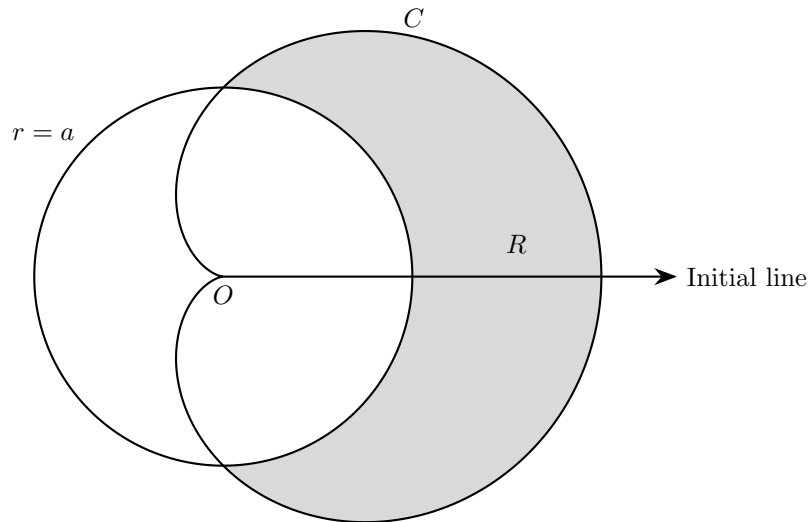
$$\int \cos^2 \theta \, d\theta = \frac{\theta}{2} + \frac{\sin 2\theta}{4}$$

Therefore

$$\begin{aligned} A &= 8a^2 \left[\frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_0^{\pi/4} - 2a^2 [\tan \theta]_0^{\pi/4} \\ &= 8a^2 \left(\frac{\pi}{8} + \frac{1}{4} \right) - 2a^2(1) \\ &= \pi a^2 + 2a^2 - 2a^2 \\ &= \pi a^2 \end{aligned}$$

Hence the shaded area is

$$\boxed{\pi a^2}$$



11. The figure above shows a sketch of the curve C with polar equation

$$r = a(1 + \cos \theta)$$

and the circle

$$r = a$$

where $a > 0$ and $-\pi \leq \theta \leq \pi$.

The shaded region inside C and outside the circle has area

$$18 + \frac{9\pi}{4}$$

Find the value of a .

[8]

Solution

For the shaded region, we first find where the curve and the circle meet.

The curve is

$$r = a(1 + \cos \theta)$$

and the circle is

$$r = a$$

At points of intersection,

$$a(1 + \cos \theta) = a$$

Since $a > 0$, divide by a :

$$1 + \cos \theta = 1$$

so

$$\cos \theta = 0$$

Hence

$$\theta = \pm \frac{\pi}{2}$$

So the shaded region is between $\theta = -\frac{\pi}{2}$ and $\theta = \frac{\pi}{2}$, with outer radius

$$r = a(1 + \cos \theta)$$

and inner radius

$$r = a$$

Using the polar area formula,

$$\text{Area} = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta$$

the shaded area is

$$\frac{1}{2} \int_{-\pi/2}^{\pi/2} ([a(1 + \cos \theta)]^2 - a^2) d\theta$$

Expand the bracket:

$$\begin{aligned} [a(1 + \cos \theta)]^2 - a^2 &= a^2(1 + 2 \cos \theta + \cos^2 \theta) - a^2 \\ &= a^2(2 \cos \theta + \cos^2 \theta) \end{aligned}$$

So

$$\text{Area} = \frac{a^2}{2} \int_{-\pi/2}^{\pi/2} (2 \cos \theta + \cos^2 \theta) d\theta$$

Now evaluate the two parts separately.

First,

$$\int_{-\pi/2}^{\pi/2} 2 \cos \theta d\theta = 2[\sin \theta]_{-\pi/2}^{\pi/2} = 2(1 - (-1)) = 4$$

For the second part, use

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

So

$$\begin{aligned} \int_{-\pi/2}^{\pi/2} \cos^2 \theta d\theta &= \int_{-\pi/2}^{\pi/2} \frac{1 + \cos 2\theta}{2} d\theta \\ &= \left[\frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_{-\pi/2}^{\pi/2} \\ &= \left(\frac{\pi}{4} + 0 \right) - \left(-\frac{\pi}{4} + 0 \right) \\ &= \frac{\pi}{2} \end{aligned}$$

Therefore

$$\begin{aligned} \text{Area} &= \frac{a^2}{2} \left(4 + \frac{\pi}{2} \right) \\ &= 2a^2 + \frac{\pi a^2}{4} \end{aligned}$$

We are told this area is

$$18 + \frac{9\pi}{4}$$

So

$$2a^2 + \frac{\pi a^2}{4} = 18 + \frac{9\pi}{4}$$

Multiply through by 4:

$$\begin{aligned} 8a^2 + \pi a^2 &= 72 + 9\pi \\ a^2(8 + \pi) &= 9(8 + \pi) \end{aligned}$$

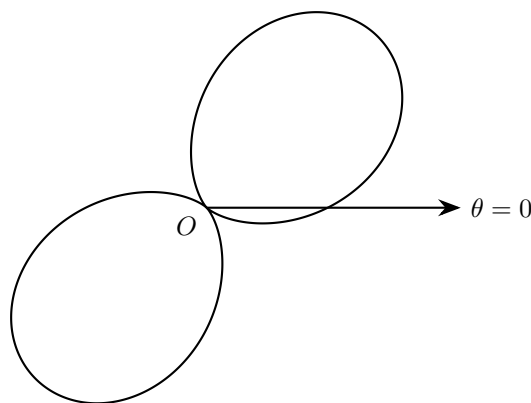
Hence

$$a^2 = 9$$

Since $a > 0$,

$$a = 3$$

Therefore, the value of a is 3.



12. The diagram below shows the curve with polar equation $r = 1 + \sin 2\theta$ for $0 \leq \theta \leq 2\pi$.

- (a) Find the values of θ at the pole. [1]
- (b) Find the polar coordinates of the points on the curve where r takes its maximum value. [2]
- (c) Find the exact total area enclosed by the curve. [4]
- (d) Find a Cartesian equation for the curve. [2]

Solution

(a) At the pole, $r = 0$. So

$$1 + \sin 2\theta = 0$$

$$\sin 2\theta = -1$$

For $0 \leq \theta \leq 2\pi$, this gives

$$2\theta = \frac{3\pi}{2}, \frac{7\pi}{2}$$

so

$$\theta = \frac{3\pi}{4}, \frac{7\pi}{4}$$

Therefore the values of θ at the pole are $\frac{3\pi}{4}$ and $\frac{7\pi}{4}$.

(b) $r = 1 + \sin 2\theta$ is largest when $\sin 2\theta = 1$. Hence

$$r_{\max} = 1 + 1 = 2$$

Now $\sin 2\theta = 1$ when

$$2\theta = \frac{\pi}{2}, \frac{5\pi}{2}$$

so

$$\theta = \frac{\pi}{4}, \frac{5\pi}{4}$$

Therefore the points are

$$(2, \frac{\pi}{4}) \quad \text{and} \quad (2, \frac{5\pi}{4})$$

(c) The total area enclosed by a polar curve is

$$A = \frac{1}{2} \int_0^{2\pi} r^2 \, d\theta$$

So here

$$A = \frac{1}{2} \int_0^{2\pi} (1 + \sin 2\theta)^2 \, d\theta$$

Expanding,

$$A = \frac{1}{2} \int_0^{2\pi} (1 + 2 \sin 2\theta + \sin^2 2\theta) \, d\theta$$

Using

$$\sin^2 2\theta = \frac{1 - \cos 4\theta}{2}$$

gives

$$\begin{aligned} A &= \frac{1}{2} \int_0^{2\pi} \left(1 + 2 \sin 2\theta + \frac{1 - \cos 4\theta}{2} \right) \, d\theta \\ &= \frac{1}{2} \int_0^{2\pi} \left(\frac{3}{2} + 2 \sin 2\theta - \frac{1}{2} \cos 4\theta \right) \, d\theta \end{aligned}$$

Integrating term by term,

$$\begin{aligned} A &= \frac{1}{2} \left[\frac{3}{2}\theta - \cos 2\theta - \frac{1}{8} \sin 4\theta \right]_0^{2\pi} \\ &= \frac{1}{2} (3\pi - 1 - 0 - (-1)) \\ &= \frac{1}{2} (3\pi) \end{aligned}$$

Hence the exact total area enclosed is

$$\frac{3\pi}{2}$$

(d) Using

$$x = r \cos \theta, \quad y = r \sin \theta, \quad r^2 = x^2 + y^2$$

and $\sin 2\theta = 2 \sin \theta \cos \theta$,

$$r = 1 + 2 \sin \theta \cos \theta$$

Now

$$\sin \theta = \frac{y}{r}, \quad \cos \theta = \frac{x}{r}$$

so

$$r = 1 + \frac{2xy}{r^2}$$

Multiplying by r^2 ,

$$r^3 = r^2 + 2xy$$

Since $r^2 = x^2 + y^2$,

$$r^3 = x^2 + y^2 + 2xy = (x + y)^2$$

Also $r = \sqrt{x^2 + y^2}$, so the cartesian equation is

$$(x^2 + y^2)^{3/2} = (x + y)^2$$